LIGHTNING TALK @ RISE26

Sparse by Nature, Parallel by Design: HPC for the Life Science

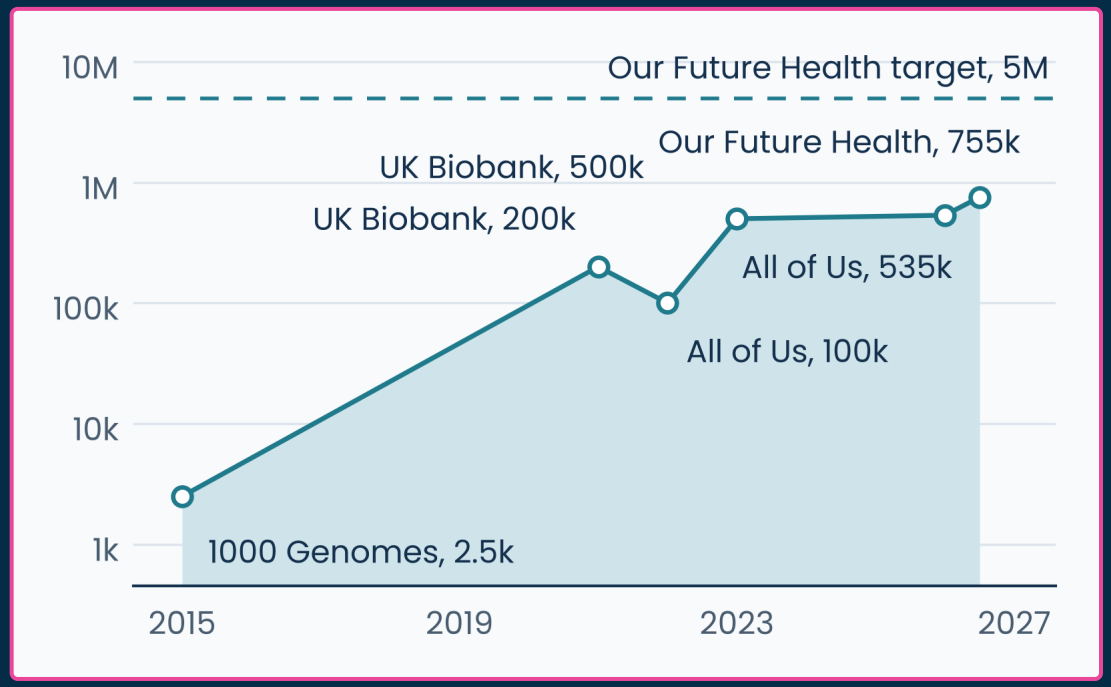
Giulia Guidi, Cornell University · gguidi@cornell.edu



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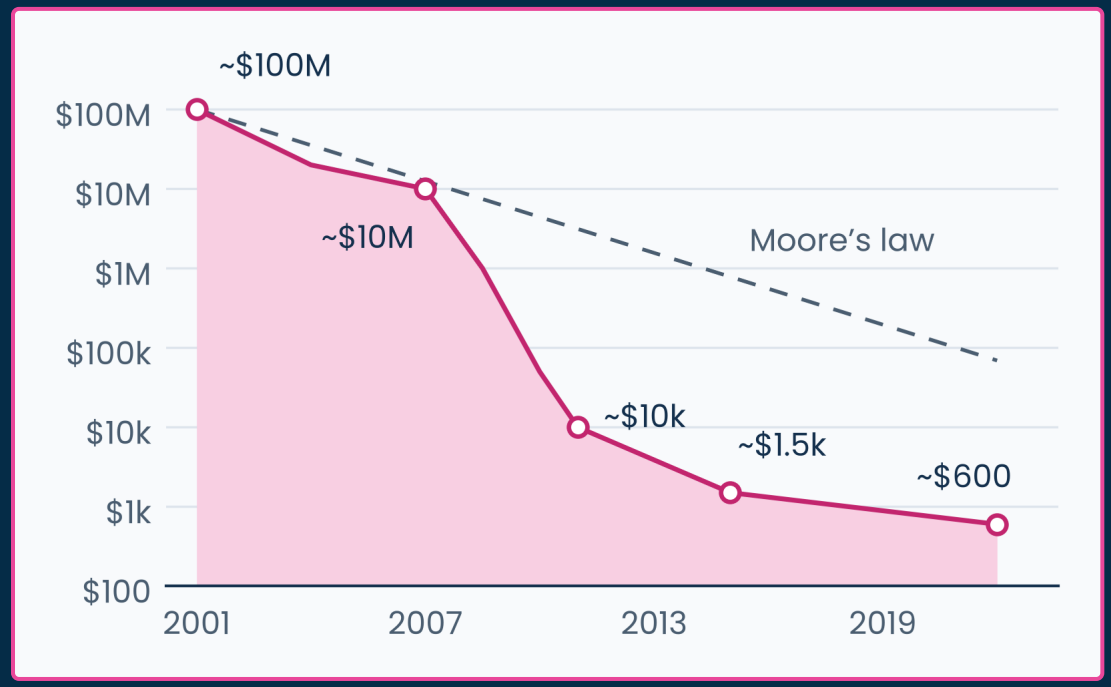
Genomics Data Outpacing Analysis Tools

1 · Participants per Cohort



1000 Genomes 2015 · UK Biobank 2021, 2023 · All of Us 2022, 2026 · Our Future Health 2026

2 · Cost per Genome



NHGRI

Data becoming orders of magnitude bigger and cheaper

Sparse Linear Algebra as Computational Paradigm

If you map your problem to (sparse) **linear algebra**, you get **performance "for free"**

KEY TAKEAWAY

Research Overview

IEEE TCHPC Early Career Award 2026

SIAG/Supercomputing Early Career Prize 2024

Gordon Bell Finalist

To be integrated in PETSc

Best Poster Award ICS26
NSF ACED Award 2024

Genome Assembly
ACDA21, IPDPS21, ICPP22, SC23

Protein Similarity Search
SC20, SC22

Kernel Clustering
PPOPP25, IPDPS26

Biobank Genomics
IA³26, Under Review Nature Computational Science

LLM Sparse Inference
Under submission

Biobank Genomics

Kernel PCA at Scale

Demographic History at Scale

Sparse Linear Algebra Primitives

SpGEMM SpMV SpTRSV SpMM SpSYRK ...

Communication-avoiding, hierarchy-aware primitives ICS26 (2), IA³26, Under Review TOMS

1 Pattern Recovery

2 Portable Compilation

3 Hardware Co-Design

CPUs | GPUs | Distributed Memory | Processing-in-Memory | AI Accelerator

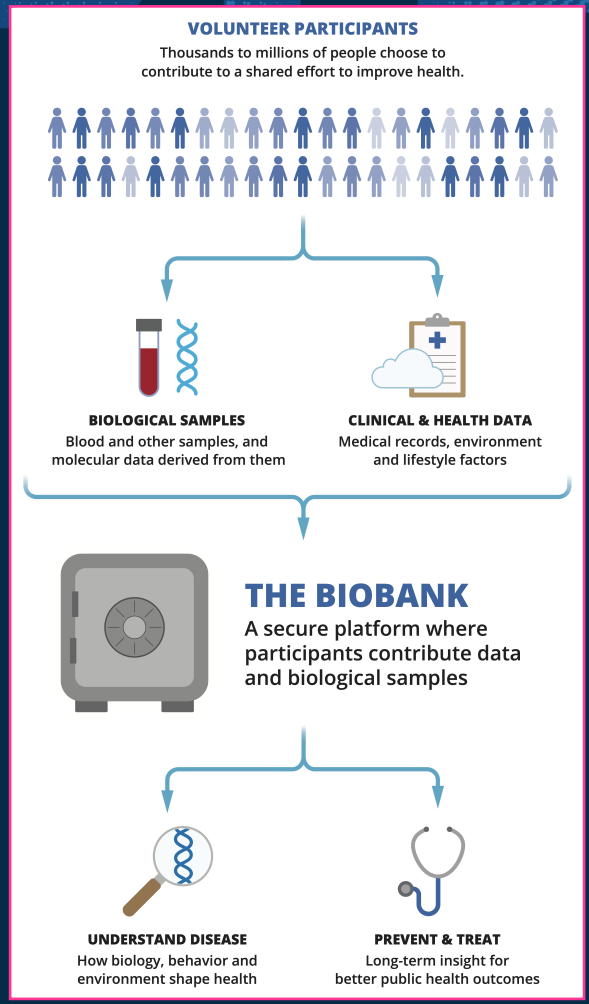
Biobank-Scale Analysis

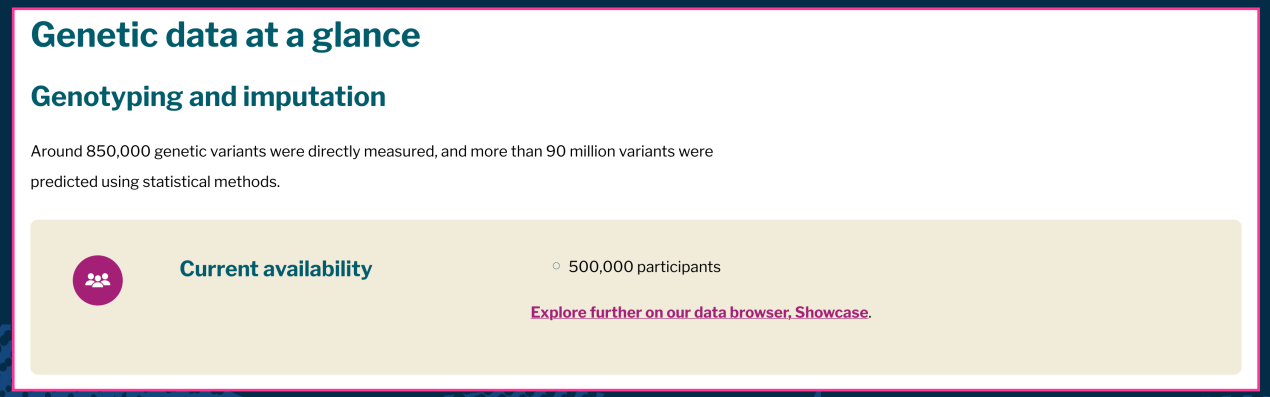
E.g., **UK Biobank and All of Us Research Program**

A biobank is a secure repository of biological samples, including **genomics samples**

Biobanks typically contain samples and data from **many thousands**, even **millions**, of individuals with and without disease, making biobanks a *valuable resource for research*

E.g., Discovery of genetic factors contributing to common and chronic diseases





Genetic data at a glance

Genotyping and imputation

Around 850,000 genetic variants were directly measured, and more than 90 million variants were predicted using statistical methods.

Current availability

- 500,000 participants

Explore further on our data browser, Showcase.

broadinstitute.org/what-is-a-biobank · ukbiobank.ac.uk/about-our-data/types-of-data/genetic-data

SpMV → Graph Traversal → SpTrSV → SpMV (Mikado)




Hey, we started with sparse matrices — **let's go back, but smarter**!

← – – – – – – – – full circle – – – – – – – – →

Genotype Matrix	GRG	GRG	GRG
SpMV	Graph Traversal	SpTrSV	k · SpMV
Dot Products over Rows	Height-based Traversal on DAG	(I-A)x = b Sparse Triangular Solve	Block Structure → k Fast SpMVs

SpMV is easier to optimize — and it already ships on many hardware platform!



Mikado Preprint (under submission at Nature Methods)
Mikado Code: https://github.com/CornellHPC/Mikado

Currently using optimized cuSPARSE and rocSPARSE to perform the SpMVs

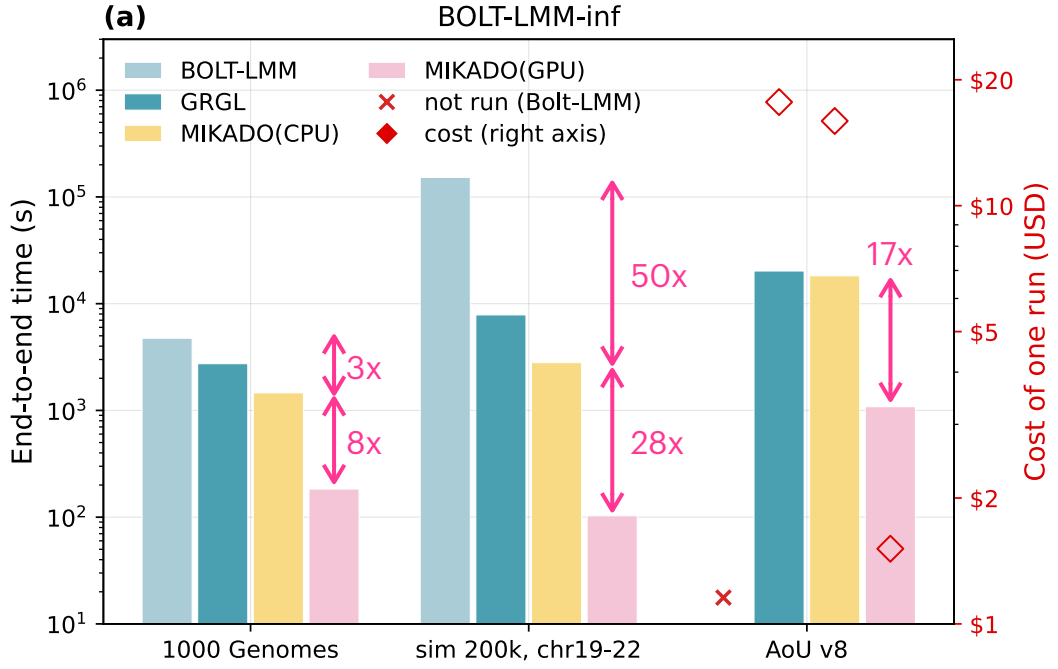
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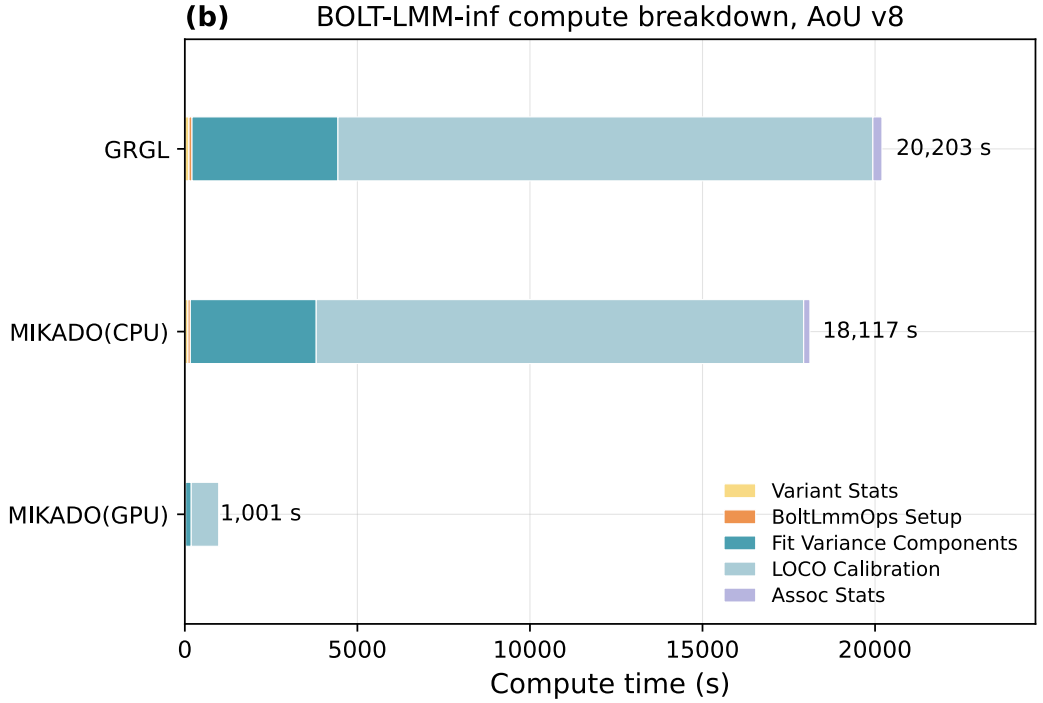
BOLT-LMM

BOLT-LMM is the standard tool for finding which variants are linked to a trait

20x faster and about **10x cheaper** than graph traversal on All of Us, where stock BOLT-LMM cannot run at all

NVIDIA A100





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If you want to learn more on our work:

1 INVITED TALK IN THE "AI IN HEALTHCARE AND CANCER RESEARCH" SESSION ON TUESDAY 11/17 11:15 AM

2 PAPER PRESENTATIONS IN THE IA3 WORKSHOP THIS AFTERNOON 11/15

4 POSTERS DURING THE POSTER SESSION ON TUESDAY 11/17 AFTERNOON

Thank You!

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